ResNet10 Training Summary

The training is ✅ **complete** ("✓ All done!"). Here's what I can confirm:

Model	Dataset	Test Accuracy	Notes
ResNet-10 (scratch)	ASL Commands (3 classes)	*(scrolled off terminal)*	Model saved ✓
ResNet-10 (scratch)	ASL Digits (10 classes)	*(scrolled off terminal)*	Model saved ✓
ResNet-10 (scratch)	ASL Alphabets (26 classes)	*(scrolled off terminal)*	Model saved ✓
ResNet-50 TL	**ASL Alphabets (26 classes)**	**96.00%**	Ran all 30 epochs

From the **Transfer Learning** classification report visible in the terminal:

- **Overall accuracy: 96.00%** (macro avg F1: 0.9599)
- Best per-class: **C** (100% F1), **D** (99.67%)
- Weakest per-class: **X** (90.24%), **V** (91.29%)

```
============================================================
 Classification Report — ResNet-10 — ASL Commands
============================================================
              precision    recall  f1-score   support

         del     1.0000    1.0000    1.0000       300
     nothing     1.0000    1.0000    1.0000       300
       space     1.0000    1.0000    1.0000       300

    accuracy                         1.0000       900
```

```
   macro avg     1.0000    1.0000    1.0000       900
weighted avg     1.0000    1.0000    1.0000       900

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\commands_confusion.png

  Test Accuracy: 1.0000

============================================================
  Classification Report — ResNet-10 — ASL Digits
============================================================
              precision    recall  f1-score   support

           0     1.0000    0.9048    0.9500        21
           1     0.9130    1.0000    0.9545        21
           2     1.0000    1.0000    1.0000        21
           3     1.0000    0.9524    0.9756        21
           4     1.0000    0.9048    0.9500        21
           5     0.9545    1.0000    0.9767        21
           6     1.0000    0.9524    0.9756        21
           7     1.0000    0.9524    0.9756        21
           8     0.9545    1.0000    0.9767        21
           9     0.8750    1.0000    0.9333        21

    accuracy                         0.9667       210
   macro avg     0.9697    0.9667    0.9668       210
```

```
weighted avg    0.9697    0.9667    0.9668      210

 Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\digits_confusion.png

 Test Accuracy: 0.9667

============================================================
 Classification Report — ResNet-10 — ASL Alphabets
============================================================
          precision    recall  f1-score   support

       A     1.0000    1.0000    1.0000       150
       B     1.0000    1.0000    1.0000       150
       C     1.0000    1.0000    1.0000       150
       D     1.0000    1.0000    1.0000       150
       E     1.0000    1.0000    1.0000       150
       F     1.0000    1.0000    1.0000       150
       G     1.0000    1.0000    1.0000       150
       H     1.0000    1.0000    1.0000       150
       I     0.9934    1.0000    0.9967       150
       J     1.0000    0.9933    0.9967       150
       K     1.0000    1.0000    1.0000       150
       L     1.0000    1.0000    1.0000       150
       M     0.9934    1.0000    0.9967       150
       N     1.0000    0.9933    0.9967       150
       O     1.0000    1.0000    1.0000       150
```

```
           P    1.0000    1.0000    1.0000       150
           Q    1.0000    1.0000    1.0000       150
           R    1.0000    1.0000    1.0000       150
           S    1.0000    1.0000    1.0000       150
           T    1.0000    1.0000    1.0000       150
           U    1.0000    1.0000    1.0000       150
           V    1.0000    1.0000    1.0000       150
           W    1.0000    1.0000    1.0000       150
           X    1.0000    1.0000    1.0000       150
           Y    1.0000    1.0000    1.0000       150
           Z    1.0000    1.0000    1.0000       150

    accuracy                         0.9995      3900
   macro avg    0.9995    0.9995    0.9995      3900
weighted avg    0.9995    0.9995    0.9995      3900

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_confusion.png

  Test Accuracy: 0.9995

============================================================
  Classification Report — ResNet-50 Transfer Learning — ASL Alphabets
============================================================
              precision    recall  f1-score   support
```

A	0.9735	0.9800	0.9767	150
B	0.9494	1.0000	0.9740	150
C	1.0000	1.0000	1.0000	150
D	0.9934	1.0000	0.9967	150
E	0.9600	0.9600	0.9600	150
F	0.9803	0.9933	0.9868	150
G	0.9866	0.9800	0.9833	150
H	0.9739	0.9933	0.9835	150
I	0.9664	0.9600	0.9632	150
J	0.9799	0.9733	0.9766	150
K	0.9801	0.9867	0.9834	150
L	0.9673	0.9867	0.9769	150
M	0.9145	0.9267	0.9205	150
N	0.9257	0.9133	0.9195	150
O	0.9933	0.9867	0.9900	150
P	1.0000	0.9533	0.9761	150
Q	0.9804	1.0000	0.9901	150
R	0.9463	0.9400	0.9431	150
S	0.8827	0.9533	0.9167	150
T	0.9632	0.8733	0.9161	150
U	0.9524	0.9333	0.9428	150
V	0.9562	0.8733	0.9129	150
W	0.9125	0.9733	0.9419	150
X	0.9116	0.8933	0.9024	150
Y	0.9603	0.9667	0.9635	150
Z	0.9600	0.9600	0.9600	150

```
    accuracy                    0.9600     3900
   macro avg    0.9604    0.9600    0.9599     3900
weighted avg    0.9604    0.9600    0.9599     3900

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_tl_confusion.png

  Test Accuracy: 0.9600
```

terminal output:

```
PS C:\Users\samyb\Downloads\ai-models> py notebooks/ResNet10_ASL.py
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  ResNet-10 — ASL Commands
============================================================
  Train: 7200 images | Val: 900 | Test: 900
  Classes: ['del', 'nothing', 'space']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
```

```
VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

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VRAM: 8.0 GB

Device: cuda

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VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

 Epoch 1/30 — loss: 0.4955 acc: 0.7588 val_loss: 0.5245 val_acc: 0.8044 [38.2s]

 Epoch 2/30 — loss: 0.1594 acc: 0.9404 val_loss: 0.2760 val_acc: 0.9100 [5.8s]

 Epoch 3/30 — loss: 0.0869 acc: 0.9682 val_loss: 0.0416 val_acc: 0.9867 [5.8s]

 Epoch 4/30 — loss: 0.0836 acc: 0.9721 val_loss: 0.0103 val_acc: 1.0000 [5.9s]
```

```
Epoch  5/30 — loss: 0.0126  acc: 0.9967  val_loss: 0.0113  val_acc: 0.9967  [5.8s]
Epoch  6/30 — loss: 0.0168  acc: 0.9953  val_loss: 0.0045  val_acc: 1.0000  [5.8s]
Epoch  7/30 — loss: 0.0079  acc: 0.9975  val_loss: 0.0002  val_acc: 1.0000  [5.8s]
Epoch  8/30 — loss: 0.0177  acc: 0.9936  val_loss: 0.3007  val_acc: 0.8900  [5.8s]
Epoch  9/30 — loss: 0.0086  acc: 0.9979  val_loss: 0.4718  val_acc: 0.9689  [5.8s]
Epoch 10/30 — loss: 0.0164  acc: 0.9949  val_loss: 0.0024  val_acc: 1.0000  [5.8s]
Epoch 11/30 — loss: 0.0030  acc: 0.9994  val_loss: 0.0001  val_acc: 1.0000  [5.8s]
Epoch 12/30 — loss: 0.0034  acc: 0.9989  val_loss: 0.0001  val_acc: 1.0000  [5.9s]
Epoch 13/30 — loss: 0.0026  acc: 0.9994  val_loss: 0.0001  val_acc: 1.0000  [5.9s]
Epoch 14/30 — loss: 0.0029  acc: 0.9993  val_loss: 0.0494  val_acc: 0.9811  [5.8s]
Epoch 15/30 — loss: 0.0277  acc: 0.9922  val_loss: 0.0095  val_acc: 0.9989  [5.7s]
Epoch 16/30 — loss: 0.0039  acc: 0.9990  val_loss: 0.0001  val_acc: 1.0000  [5.6s]
Epoch 17/30 — loss: 0.0008  acc: 1.0000  val_loss: 0.0001  val_acc: 1.0000  [5.6s]
Epoch 18/30 — loss: 0.0004  acc: 1.0000  val_loss: 0.0015  val_acc: 0.9989  [5.7s]
Epoch 19/30 — loss: 0.0007  acc: 0.9997  val_loss: 0.0000  val_acc: 1.0000  [5.6s]
Epoch 20/30 — loss: 0.0002  acc: 1.0000  val_loss: 0.0000  val_acc: 1.0000  [5.7s]
Epoch 21/30 — loss: 0.0149  acc: 0.9960  val_loss: 0.4460  val_acc: 0.9333  [5.7s]
Epoch 22/30 — loss: 0.0108  acc: 0.9965  val_loss: 0.0001  val_acc: 1.0000  [5.6s]
Epoch 23/30 — loss: 0.0005  acc: 0.9999  val_loss: 0.0000  val_acc: 1.0000  [5.7s]
Epoch 24/30 — loss: 0.0006  acc: 1.0000  val_loss: 0.0000  val_acc: 1.0000  [5.6s]
Epoch 25/30 — loss: 0.0002  acc: 1.0000  val_loss: 0.0000  val_acc: 1.0000  [5.7s]
Epoch 26/30 — loss: 0.0003  acc: 1.0000  val_loss: 0.0002  val_acc: 1.0000  [5.7s]
Epoch 27/30 — loss: 0.0004  acc: 1.0000  val_loss: 0.0000  val_acc: 1.0000  [5.6s]
Epoch 28/30 — loss: 0.0062  acc: 0.9978  val_loss: 0.0205  val_acc: 0.9944  [5.6s]
Epoch 29/30 — loss: 0.0324  acc: 0.9883  val_loss: 0.0034  val_acc: 1.0000  [5.6s]
Epoch 30/30 — loss: 0.0127  acc: 0.9972  val_loss: 0.0001  val_acc: 1.0000  [5.7s]
```

```
  Early stopping at epoch 30 (patience=5)
  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\commands_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  Classification Report — ResNet-10 — ASL Commands
============================================================
              precision    recall  f1-score   support

         del     1.0000    1.0000    1.0000       300
     nothing     1.0000    1.0000    1.0000       300
       space     1.0000    1.0000    1.0000       300

    accuracy                         1.0000       900
   macro avg     1.0000    1.0000    1.0000       900
```

```
weighted avg     1.0000    1.0000    1.0000      900

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\commands_confusion.png

  Test Accuracy: 1.0000
  ✓ Model saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\commands_model.pth

============================================================
  ResNet-10 — ASL Digits
============================================================
  Train: 1644 images | Val: 208 | Test: 210
  Classes: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
```

```
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
 Epoch  1/30 — loss: 1.9056  acc: 0.3187  val_loss: 5.5642  val_acc: 0.1154  [32.4s]
 Epoch  2/30 — loss: 1.1314  acc: 0.6210  val_loss: 1.2472  val_acc: 0.5144  [1.6s]
 Epoch  3/30 — loss: 0.7110  acc: 0.7816  val_loss: 0.9925  val_acc: 0.6250  [1.5s]
 Epoch  4/30 — loss: 0.4481  acc: 0.8844  val_loss: 1.0795  val_acc: 0.5817  [1.6s]
 Epoch  5/30 — loss: 0.3093  acc: 0.9179  val_loss: 0.9466  val_acc: 0.6490  [1.6s]
 Epoch  6/30 — loss: 0.3386  acc: 0.8978  val_loss: 0.3907  val_acc: 0.8846  [1.6s]
 Epoch  7/30 — loss: 0.2106  acc: 0.9519  val_loss: 0.2508  val_acc: 0.9375  [1.6s]
 Epoch  8/30 — loss: 0.1674  acc: 0.9568  val_loss: 0.1398  val_acc: 0.9615  [1.9s]
 Epoch  9/30 — loss: 0.1570  acc: 0.9599  val_loss: 1.3711  val_acc: 0.7308  [1.9s]
 Epoch 10/30 — loss: 0.1426  acc: 0.9605  val_loss: 0.2631  val_acc: 0.9135  [1.6s]
 Epoch 11/30 — loss: 0.1113  acc: 0.9665  val_loss: 0.2629  val_acc: 0.9087  [1.5s]
 Epoch 12/30 — loss: 0.1105  acc: 0.9696  val_loss: 1.1055  val_acc: 0.6731  [1.6s]
 Epoch 13/30 — loss: 0.1192  acc: 0.9653  val_loss: 0.2921  val_acc: 0.9087  [1.6s]
 Early stopping at epoch 13 (patience=5)
 Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\digits_history.png
```

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

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Classification Report — ResNet-10 — ASL Digits

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	precision	recall	f1-score	support
0	1.0000	0.9048	0.9500	21
1	0.9130	1.0000	0.9545	21
2	1.0000	1.0000	1.0000	21
3	1.0000	0.9524	0.9756	21
4	1.0000	0.9048	0.9500	21
5	0.9545	1.0000	0.9767	21
6	1.0000	0.9524	0.9756	21
7	1.0000	0.9524	0.9756	21

```
           8    0.9545    1.0000    0.9767        21
           9    0.8750    1.0000    0.9333        21

    accuracy                         0.9667       210
   macro avg    0.9697    0.9667    0.9668       210
weighted avg    0.9697    0.9667    0.9668       210

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\digits_confusion.png

  Test Accuracy: 0.9667
  ✓ Model saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\digits_model.pth

============================================================
  ResNet-10 — ASL Alphabets
============================================================
  Train: 31200 images | Val: 3900 | Test: 3900
  Classes: ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
```

```
VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

 Epoch  1/30 ━━━━━━━━━━━━━━━━━━━━ loss: 1.4254  acc: 0.5455  val_loss: 0.4284  val_acc: 0.8279  [55.4s]

 Epoch  2/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.2420  acc: 0.9271  val_loss: 0.1289  val_acc: 0.9497  [24.2s]

 Epoch  3/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.1079  acc: 0.9685  val_loss: 0.1063  val_acc: 0.9677  [24.2s]

 Epoch  4/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0731  acc: 0.9783  val_loss: 0.0556  val_acc: 0.9797  [24.9s]

 Epoch  5/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0474  acc: 0.9864  val_loss: 0.0271  val_acc: 0.9910  [24.8s]

 Epoch  6/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0471  acc: 0.9857  val_loss: 0.0643  val_acc: 0.9785  [24.7s]

 Epoch  7/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0385  acc: 0.9873  val_loss: 0.0895  val_acc: 0.9672  [24.6s]

 Epoch  8/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0371  acc: 0.9883  val_loss: 0.0106  val_acc: 0.9954  [24.3s]

 Epoch  9/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0312  acc: 0.9898  val_loss: 0.0192  val_acc: 0.9941  [24.3s]

 Epoch 10/30 ━━━━━━━━━━━━━━━━━━━━ loss: 0.0334  acc: 0.9897  val_loss: 0.1519  val_acc: 0.9449  [24.3s]
```

```
  Epoch 11/30 — loss: 0.0234  acc: 0.9928  val_loss: 0.0072  val_acc: 0.9979  [24.3s]
  Epoch 12/30 — loss: 0.0302  acc: 0.9901  val_loss: 0.0201  val_acc: 0.9936  [24.5s]
  Epoch 13/30 — loss: 0.0245  acc: 0.9918  val_loss: 0.0054  val_acc: 0.9985  [24.3s]
  Epoch 14/30 — loss: 0.0184  acc: 0.9938  val_loss: 0.0089  val_acc: 0.9962  [24.2s]
  Epoch 15/30 — loss: 0.0244  acc: 0.9919  val_loss: 0.0124  val_acc: 0.9949  [24.3s]
  Epoch 16/30 — loss: 0.0171  acc: 0.9947  val_loss: 0.0058  val_acc: 0.9982  [24.4s]
  Epoch 17/30 — loss: 0.0185  acc: 0.9941  val_loss: 0.0060  val_acc: 0.9974  [24.2s]
  Epoch 18/30 — loss: 0.0107  acc: 0.9971  val_loss: 0.0057  val_acc: 0.9985  [25.3s]
  Early stopping at epoch 18 (patience=5)
  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  Classification Report — ResNet-10 — ASL Alphabets
============================================================
```

	precision	recall	f1-score	support
A	1.0000	1.0000	1.0000	150
B	1.0000	1.0000	1.0000	150
C	1.0000	1.0000	1.0000	150
D	1.0000	1.0000	1.0000	150
E	1.0000	1.0000	1.0000	150
F	1.0000	1.0000	1.0000	150
G	1.0000	1.0000	1.0000	150
H	1.0000	1.0000	1.0000	150
I	0.9934	1.0000	0.9967	150
J	1.0000	0.9933	0.9967	150
K	1.0000	1.0000	1.0000	150
L	1.0000	1.0000	1.0000	150
M	0.9934	1.0000	0.9967	150
N	1.0000	0.9933	0.9967	150
O	1.0000	1.0000	1.0000	150
P	1.0000	1.0000	1.0000	150
Q	1.0000	1.0000	1.0000	150
R	1.0000	1.0000	1.0000	150
S	1.0000	1.0000	1.0000	150
T	1.0000	1.0000	1.0000	150
U	1.0000	1.0000	1.0000	150
V	1.0000	1.0000	1.0000	150
W	1.0000	1.0000	1.0000	150
X	1.0000	1.0000	1.0000	150

```
           Y    1.0000    1.0000    1.0000       150
           Z    1.0000    1.0000    1.0000       150

    accuracy                         0.9995      3900
   macro avg    0.9995    0.9995    0.9995      3900
weighted avg    0.9995    0.9995    0.9995      3900

  Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_confusion.png

  Test Accuracy: 0.9995
  ✓ Model saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_model.pth

============================================================
  ResNet-50 Transfer Learning — ASL Alphabets
============================================================
  Train: 31200 images | Val: 3900 | Test: 3900
  Classes: ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z']
Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to C:\Users\samyb/.cache\torch\hub\checkpoints\resnet50-11ad3fa6.pth
100%|[redacted]| 97.8M/97.8M [00:12<00:00, 8.21MB/s]
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
```

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Epoch 1/30 — loss: 1.6322 acc: 0.5339 val_loss: 0.6317 val_acc: 0.8215 [69.0s]
Epoch 2/30 — loss: 0.8803 acc: 0.7220 val_loss: 0.4102 val_acc: 0.8749 [37.8s]
Epoch 3/30 — loss: 0.7434 acc: 0.7589 val_loss: 0.3262 val_acc: 0.9013 [38.0s]
Epoch 4/30 — loss: 0.6766 acc: 0.7791 val_loss: 0.3037 val_acc: 0.9105 [38.1s]
Epoch 5/30 — loss: 0.6360 acc: 0.7928 val_loss: 0.2777 val_acc: 0.9167 [37.8s]
```

```
Epoch  6/30 — loss: 0.6082  acc: 0.8006  val_loss: 0.2355  val_acc: 0.9256  [37.8s]
Epoch  7/30 — loss: 0.5939  acc: 0.8051  val_loss: 0.2363  val_acc: 0.9323  [36.9s]
Epoch  8/30 — loss: 0.5751  acc: 0.8090  val_loss: 0.2152  val_acc: 0.9349  [36.7s]
Epoch  9/30 — loss: 0.5647  acc: 0.8146  val_loss: 0.2139  val_acc: 0.9318  [36.6s]
Epoch 10/30 — loss: 0.5474  acc: 0.8195  val_loss: 0.2009  val_acc: 0.9374  [37.5s]
Epoch 11/30 — loss: 0.5351  acc: 0.8221  val_loss: 0.1993  val_acc: 0.9418  [37.8s]
Epoch 12/30 — loss: 0.5249  acc: 0.8274  val_loss: 0.1981  val_acc: 0.9392  [37.1s]
Epoch 13/30 — loss: 0.5203  acc: 0.8263  val_loss: 0.1825  val_acc: 0.9451  [36.6s]
Epoch 14/30 — loss: 0.5125  acc: 0.8288  val_loss: 0.1805  val_acc: 0.9426  [36.5s]
Epoch 15/30 — loss: 0.4983  acc: 0.8306  val_loss: 0.1675  val_acc: 0.9456  [36.5s]
Epoch 16/30 — loss: 0.4952  acc: 0.8328  val_loss: 0.1632  val_acc: 0.9485  [36.8s]
Epoch 17/30 — loss: 0.4842  acc: 0.8372  val_loss: 0.1542  val_acc: 0.9497  [37.7s]
Epoch 18/30 — loss: 0.4826  acc: 0.8370  val_loss: 0.1529  val_acc: 0.9546  [37.9s]
Epoch 19/30 — loss: 0.4801  acc: 0.8388  val_loss: 0.1654  val_acc: 0.9492  [37.8s]
Epoch 20/30 — loss: 0.4774  acc: 0.8416  val_loss: 0.1528  val_acc: 0.9510  [37.8s]
Epoch 21/30 — loss: 0.4627  acc: 0.8450  val_loss: 0.1512  val_acc: 0.9528  [37.7s]
Epoch 22/30 — loss: 0.4667  acc: 0.8415  val_loss: 0.1470  val_acc: 0.9562  [37.8s]
Epoch 23/30 — loss: 0.4539  acc: 0.8486  val_loss: 0.1519  val_acc: 0.9510  [37.8s]
Epoch 24/30 — loss: 0.4562  acc: 0.8469  val_loss: 0.1470  val_acc: 0.9538  [37.9s]
Epoch 25/30 — loss: 0.4485  acc: 0.8501  val_loss: 0.1469  val_acc: 0.9536  [37.8s]
Epoch 26/30 — loss: 0.4467  acc: 0.8487  val_loss: 0.1438  val_acc: 0.9559  [37.8s]
Epoch 27/30 — loss: 0.4493  acc: 0.8492  val_loss: 0.1383  val_acc: 0.9567  [37.9s]
Epoch 28/30 — loss: 0.4411  acc: 0.8525  val_loss: 0.1395  val_acc: 0.9562  [37.9s]
Epoch 29/30 — loss: 0.4445  acc: 0.8509  val_loss: 0.1348  val_acc: 0.9567  [37.7s]
Epoch 30/30 — loss: 0.4403  acc: 0.8511  val_loss: 0.1299  val_acc: 0.9633  [38.0s]
Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_tl_history.png
```

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

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Classification Report — ResNet-50 Transfer Learning — ASL Alphabets

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	precision	recall	f1-score	support
A	0.9735	0.9800	0.9767	150
B	0.9494	1.0000	0.9740	150
C	1.0000	1.0000	1.0000	150
D	0.9934	1.0000	0.9967	150
E	0.9600	0.9600	0.9600	150
F	0.9803	0.9933	0.9868	150
G	0.9866	0.9800	0.9833	150
H	0.9739	0.9933	0.9835	150

I	0.9664	0.9600	0.9632	150
J	0.9799	0.9733	0.9766	150
K	0.9801	0.9867	0.9834	150
L	0.9673	0.9867	0.9769	150
M	0.9145	0.9267	0.9205	150
N	0.9257	0.9133	0.9195	150
O	0.9933	0.9867	0.9900	150
P	1.0000	0.9533	0.9761	150
Q	0.9804	1.0000	0.9901	150
R	0.9463	0.9400	0.9431	150
S	0.8827	0.9533	0.9167	150
T	0.9632	0.8733	0.9161	150
U	0.9524	0.9333	0.9428	150
V	0.9562	0.8733	0.9129	150
W	0.9125	0.9733	0.9419	150
X	0.9116	0.8933	0.9024	150
Y	0.9603	0.9667	0.9635	150
Z	0.9600	0.9600	0.9600	150
accuracy			0.9600	3900
macro avg	0.9604	0.9600	0.9599	3900
weighted avg	0.9604	0.9600	0.9599	3900

Saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_tl_confusion.png

Test Accuracy: 0.9600

✓ Model saved: C:\Users\samyb\Downloads\ai-models\results\resnet10\alphabets_tl_model.pth

✓ All done! Results saved to: C:\Users\samyb\Downloads\ai-models\results\resnet10